

Education

City University of Hong Kong	Hong Kong SAR
B. E. in Computer and Data Engineering, Department of Electrical Engineering	September 2019 - June 2025
<i>Coursework: Cloud Computing Systems, Database Systems, Big Data, and Human-Computer Interaction</i>	
Vanderbilt University	Nashville, TN, USA
Study abroad coursework in the School of Engineering	January 2024 - May 2024
<i>Coursework: Applied Machine Learning, Artificial Intelligence, FPGA Design, and Programming Languages</i>	

Technical Skills

Programming Languages	Python, JavaScript/TypeScript, C/C++, Rust
Frameworks and Tools	Frontend/Mobile: React, Next.js, React Native, Unit Integration, E2E Testing Backend and Database: NestJS, FastAPI, PostgreSQL DevOps & Infrastructure: Git, CI/CD (GitHub Actions), AWS, Docker

Professional Experience

URJOB	Seoul, South Korea
<i>Frontend Engineering Intern</i>	March 2026 - April 2026

- Monorepo Migration: React + React Router Legacy to TanStack Start
- Separated user app and admin dashboard into a pnpm workspace monorepo, sharing runtime and design tokens
 - Established project conventions (file-based routing) and an agentic coding harness for the new stack
 - Replaced localStorage-backed JWT with HttpOnly + SameSite: Lax cookies, closing the XSS token-theft path
 - Adopted TanStack Query as the session single source of truth and defined the access-token refresh policy
- Developer Experience Improvements
- Generated TypeScript types from the backend Spring OpenAPI spec via openapi-typescript, catching contract drift at compile time
 - Centralized TypeScript versioning via pnpm catalog and unified query keys with a Query Factory pattern
 - Enforced pre-commit/push formatting and lint (oxfmt/oxlint) and commit conventions (Commitlint) via Lefthook
- Deployment Pipeline & Infrastructure
- Designed a reusable workflow_call-based GitHub Actions architecture separating CI (lint/test), Build, and Deploy across develop, preview, and production environments
 - Moved Dockerfile-based EC2 builds into GitHub Actions, shipping only the .output artifact to minimize runtime image size and remove instance build load

MotoNerv Limited	Hong Kong SAR
<i>Hardware (Embedded Systems) Engineering Intern</i>	June 2023 - August 2023
• Developed a prototype of an MCU-based (ESP32) dashcam for streaming video and collecting sensor data • Preprocessed on-car sensor and location data tailored for AI video analytics in the AWS cloud environment • Implemented data pipeline to stream sensor and video data directly to AWS S3 buckets for further processing via cellular network connection	

Projects

RunFit	 
<i>Running crew matching platform with location-based filtering and pace-based discovery</i>	
• Designed a domain-based REST API structure with typed DTO request/response bodies, standardizing the frontend-backend interface contract across a 4-person team through iterative requirement changes • Architected a Next.js server proxy layer to centralize API request/response control, preventing backend endpoint exposure and closing JWT token-handling vulnerabilities across App Router server/client component boundaries • Implemented role-based access control (RBAC) with httpOnly cookie-backed JWT, addressing secure session state across server and client components in Next.js App Router • Integrated TanStack Query with custom factory patterns for queries/mutations, establishing a structured error handling strategy across all data fetching layers	

Squash Motion Tracking (Final Year Project)



Participated in Hong Kong Sport Institute (HKSI) initiated computer vision project for athlete performance analysis

- Architected a full-stack computer-vision pipeline (FastAPI + React) for player tracking and pose detection from match video, enabling data-driven coaching
- Orchestrated SAM2 and YOLO model inference behind a REST API for video processing
- Reduced VRAM during transformer-based segmentation via frame sampling on forward/backward propagation
- Containerized the system with Docker / Docker Compose for reproducible deployment with NVIDIA GPU acceleration

Leadership Activities & Other Experiences

Frontend Short-Term Intensive Course - codeit

November 2025 - January 2026

- Completed 10-week intensive covering Jest/React Testing Library, CI/CD pipelines, and Next.js architecture patterns
- Established team workflows by defining code conventions, PR templates, and review processes for a 4-person team
- Led architecture decisions for API structure, auth flows, and state management in a production-grade running crew platform

Data Science Coaching Study: 2024 - boostcourse

July - August 2024

- Completed 4-week program on Python data analysis covering pandas, numpy, and visualization libs (matplotlib, seaborn, plotly)
- Practiced EDA techniques including correlation analysis, pivot tables, and statistical visualization on real-world KOSIS datasets

Student Resident (Senior Resident Tutor)

July 2024 - July 2025

- Provided academic mentoring and pastoral support to resident students; coordinated hall community events and welfare programs

Student Resident Cabinet (General Officer)

October 2022 - May 2023

- Contributed to student welfare initiatives and community programs as a student government officer

Military service

August 2020 - February 2022

- Completed mandatory Republic of Korea military service

Private Tutoring

- Taught International Baccalaureate (IB) Diploma Mathematics and Electrical Engineering to secondary and university-level students